



Controlled Impedance Design

From dielectric stackup to controlled-Z trace — making PCBs behave at GHz

The science behind 50 Ω , 100 Ω , and everything in between

Audience: Intermediate Engineering Students

Microstrip • Stripline • Calculation • Stackup • Manufacturing



Why Controlled Impedance Matters

When a wire stops being a wire

Reflections

Impedance mismatch → reflections
→ signal distortion + bit errors

Standards

USB / HDMI / Ethernet / DDR
ALL specify exact Z_0 values

EMI

Bad termination → ringing
→ harmonics → EMI failure

Yield

Variable impedance across
production → intermittent failures

Cost

Controlled-Z adds 10-30%
to PCB fab cost

Documentation

Required spec in
fabrication notes

Trace Topologies

Microstrip vs Stripline vs CPW

- ▶ Microstrip — trace on outer layer, ground plane below
 - ▶ Stripline — trace on inner layer, ground planes both above and below
 - ▶ Coplanar waveguide (CPW) — trace + ground on same layer
 - ▶ Edge-coupled differential pair — two traces, ground below
 - ▶ Embedded microstrip — trace below thin dielectric, ground further below
 - ▶ Asymmetric stripline — different distances to top/bottom planes
- ▶ Microstrip: easier to probe, faster to design, radiates more
 - ▶ Stripline: better SI, fully shielded, harder to debug
 - ▶ CPW: used in RF / antenna ground transitions
 - ▶ Differential pairs (any type): for USB, HDMI, Ethernet, DDR strobes
 - ▶ Most common: 50 Ω single-ended microstrip on top layer
 - ▶ For diff pairs: 90/100 Ω diff on outer layer adjacent to GND

Impedance Physics — Where Z_0 Comes From

Distributed L and C of the trace structure

- ▶ $Z_0 = \sqrt{L/C}$ where L and C are per unit length of the transmission line
- ▶ Trace L per unit length: increases with separation from ground (taller trace stack)
- ▶ Trace C per unit length: increases with trace width AND closer ground spacing
- ▶ Wider trace → lower Z_0 (more C)
- ▶ Trace further from ground → higher Z_0 (less C, more L)
- ▶ Dielectric thickness drives separation from ground reference
- ▶ Dielectric constant (Dk) — higher Dk reduces wavelength + Z_0
- ▶ FR-4 typical Dk ≈ 4.3 , Rogers RO4350B Dk ≈ 3.48 , PTFE Dk ≈ 2.2

Common Impedance Targets

What major standards require

Standard	Single-ended Z_0	Differential Z_0	Tolerance
USB 2.0 / 3.0 / USB-C	—	90 Ω diff	$\pm 10\%$
HDMI 1.4 / 2.0 / 2.1	—	100 Ω diff	$\pm 10\%$
Gigabit Ethernet (10/100/1G)	—	100 Ω diff	$\pm 15\%$
PCIe 1.x / 2.x / 3.x	—	85 Ω diff	$\pm 10\%$
DDR3 (DQ, DQS, addr)	40-50 Ω	100 Ω diff	$\pm 10\%$
DDR4	40 Ω	85-100 Ω diff	$\pm 10\%$
LVDS	—	100 Ω diff	$\pm 10\%$
MIPI CSI-2 / DSI	—	100 Ω diff	$\pm 10\%$
RF antenna feed	50 Ω	—	$\pm 5\%$
DDR4 / DDR5 reference	—	85 Ω diff	$\pm 10\%$

Stackup Design for Controlled Impedance

Trace width is calculated FROM the stackup, not chosen first

- ▶ Pick stackup FIRST — number of layers, layer thicknesses, dielectric (D_k)
- ▶ Fab publishes the stackup options (JLCPCB JLC04161H, etc.)
- ▶ Trace width is then computed from stackup + target impedance
- ▶ Example JLCPCB JLC04161H 4-layer stackup:
 - ▶ L1 (outer): 0.46 mm to L2 (GND) → 50 Ω microstrip $\approx$ 10 mil trace
 - ▶ L3 (PWR): 0.42 mm to L4 (outer) → 50 Ω microstrip $\approx$ 10 mil trace
- ▶ Stackup matters more than trace width — change stackup, all trace widths change
- ▶ Fab adjusts trace width during manufacturing to hit target Z_o within $\pm 10\%$

Impedance Calculators

Don't hand-calc unless you must

KiCad built-in

Tools → Calculator Tools
→ Microstrip / Stripline

Saturn PCB Toolkit

Free Windows tool
More options than KiCad

Polar Si9000

Industry standard
Licensed, \$\$\$

AppCAD (Keysight)

Free, microwave-oriented
Good for RF

Online calculators

Pasternack, JLCPCB, EEWeb
Free, good enough

Fab-provided

Many fabs (JLCPCB) publish
their stackup + trace tables

Worked Example — 50 Ω Microstrip on FR-4

From stackup to trace width

Goal: 50 Ω single-ended microstrip on top layer of a 4-layer FR-4 board

- ▶ Stackup: 4-layer JLCPCB JLC04161H
- ▶ L1 (signal): 1 oz Cu (35 μm)
- ▶ Prepreg to L2: 0.21 mm
- ▶ L2: GND plane (reference)
- ▶ FR-4 core: 0.36 mm
- ▶ L3: PWR plane
- ▶ Prepreg to L4: 0.21 mm
- ▶ L4 (signal): 1 oz Cu
- ▶ Dielectric: FR-4, $D_k \approx 4.4$
- ▶ Calculator input: $H = 0.21$ mm, $T = 0.035$ mm, $\epsilon_r = 4.4$, target $Z_o = 50$ Ω
- ▶ Calculator output: trace width $W \approx 0.38$ mm (~15 mil)
- ▶ Fab will trim trace width to $\pm 10\%$ during production to hit 50 Ω

Differential Pair Impedance

Two traces — coupled together

- ▶ Diff impedance Z_{diff} measured between the two traces (odd mode)
- ▶ Common-mode impedance Z_{common} measured to ground (even mode)
- ▶ Tight coupling (trace-to-trace gap = trace width) keeps Z_{diff} stable
- ▶ Loose coupling (gap > 2× trace width) approaches 2× single-ended Z_0
- ▶ For USB diff @ 90 Ω : typical 7 mil traces with 7 mil gap on 4-layer JLC04161H
- ▶ For HDMI / Ethernet @ 100 Ω : typical 6 mil traces with 8 mil gap
- ▶ For PCIe @ 85 Ω : typical 8 mil traces with 7 mil gap
- ▶ Asymmetric routing (different distances to ground) ruins Z_{diff}

Manufacturing Process for Controlled Z

What the fab does

- ▶ Step 1: Fab quotes a stackup that hits your target impedance
- ▶ Step 2: Phototool generates trace at SLIGHTLY ADJUSTED width (compensates for etch loss)
- ▶ Step 3: Etch — over-etching reduces actual trace width by 5-15%
- ▶ Step 4: Lamination — dielectric thickness can vary by $\pm 5\%$
- ▶ Step 5: Test coupons on the panel measured with TDR (Time Domain Reflectometry)
- ▶ Step 6: If outside $\pm 10\%$ tolerance, panel reworked or scrapped
- ▶ Step 7: Report attached to delivered boards (impedance specification report)

Common Mistakes

Why controlled Z is harder than it looks

- ▶ Designing trace width without specifying stackup — fab can't match Z_0
- ▶ Mismatched stackup: your calculation assumed 0.2 mm prepreg, fab uses 0.3 mm
- ▶ Diff pair not tightly coupled — Z_{diff} varies along trace
- ▶ Width transitions (chip pad → trace width) create impedance discontinuities
- ▶ Vias not optimized for the transition (anti-pad too small/big)
- ▶ Reference plane changes (signal hops from L1 to L4 — Z_{diff} changes)
- ▶ Not requesting impedance test coupon — no proof at QA
- ▶ Wrong Dk assumption — FR-4 Dk varies 4.0-4.7 between fabs

Reference Plane Discipline

Every controlled-Z trace needs a clean reference

- ▶ Controlled-Z trace = signal + reference plane SOMEWHERE
- ▶ Microstrip → reference is ground plane DIRECTLY BELOW
- ▶ Stripline → references are ground planes ABOVE AND BELOW
- ▶ Reference must be UNINTERRUPTED across the entire trace path
- ▶ Any gap, split, or via crossing breaks the impedance
- ▶ When changing layers, also change reference planes via stitching vias
- ▶ BGA escape routing: signals on inner layers — confirm continuous reference
- ▶ Star ground / split-ground designs MUST avoid crossing the split with controlled-Z traces

Cost & Quote Impact

Controlled-Z adds 10-30% to fab cost

- ▶ Standard PCB: fab uses default trace widths, no impedance guarantee
- ▶ Controlled-Z PCB: fab measures + reports actual impedance, $\pm 10\%$ tolerance
- ▶ Cost premium: 10-30% over standard (varies by fab + complexity)
- ▶ Lead time: +1-2 days (impedance test + verification)
- ▶ Specify in fab quote: 'Controlled impedance: 50 Ω single-ended, 100 Ω differential'
- ▶ Request: TDR test coupon report + delivered with boards
- ▶ Quote tighter tolerance ($\pm 5\%$) only if your design truly needs it (RF)
- ▶ Typical JLCPCB controlled-Z addition: $\sim \$30 + \$5/\text{board}$

Best Practices

Habits for production-grade controlled Z

- ▶ Pick stackup early — design rules follow
 - ▶ Use fab-published stackup options if possible
 - ▶ Trace width calculator → start point, fab adjusts
 - ▶ Tight differential pair coupling
 - ▶ Continuous reference plane under controlled traces
 - ▶ Stitching vias when layer-changing controlled traces
- ▶ Specify controlled-Z in fab quote
 - ▶ Request TDR test report
 - ▶ Document impedance targets per net group
 - ▶ Avoid abrupt width changes (use tapers)
 - ▶ Anti-pad optimization on via transitions
 - ▶ Validate with TDR / VNA on prototype

Key Takeaways

Key takeaways from this lecture

Z₀ from L and C

Distributed inductance + capacitance.

Stackup decides

Pick stackup first, calc widths after.

Reference plane sacred

Continuous + unbroken.

Diff pair tight coupling

Width = gap for stable Z_{diff}.

Fab partnership

Use their stackup, their test coupon.

Validate with TDR

Don't assume — measure.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ Eric Bogatin — 'Signal & Power Integrity Simplified' (definitive reference)
- ▶ Howard Johnson — 'High-Speed Digital Design'
- ▶ Saturn PCB Toolkit — free impedance calculator
- ▶ Polar SI9000 — industry-standard calculator
- ▶ JLCPCB / PCBWay impedance specifications + capability
- ▶ Phil's Lab YouTube — controlled-impedance walkthroughs
- ▶ Robert Feranec — DDR/USB layout videos
- ▶ IPC-2141 / IPC-2251 — controlled-impedance standards

Thank you • Build something this week.